

## Requirements Engineering (RE)

### Definition:

Requirements Engineering is the process of identifying, analyzing, documenting, and managing the requirements of a software system.

It ensures we build **the right system** according to customer needs.

### Feasibility Study

#### ✓ Meaning:

A study to check whether the proposed system is **practical and achievable**.

#### ✓ Purpose:

- Decide whether to proceed with the project.
- Identify risks and constraints.

#### ✓ Types of Feasibility:

1. **Technical Feasibility**
  - Do we have required technology and skills?
2. **Economic Feasibility**
  - Is the project cost-effective?
3. **Operational Feasibility**
  - Will users accept the system?
4. **Schedule Feasibility**
  - Can it be completed on time?

#### ✓ Output:

Feasibility Report

## 2 Requirements Elicitation & Analysis

#### ✓ Meaning:

Process of **gathering and understanding requirements** from stakeholders.

#### ✓ Elicitation Techniques:

- Interviews
- Questionnaires
- Observation

- Workshops
- Prototyping

#### ✓ Analysis:

- Identify conflicts
- Remove ambiguity
- Prioritize requirements
- Classify requirements

#### ✓ Types of Requirements:

1. **Functional Requirements**
  - What system should do
  - Example: “System should allow login.”
2. **Non-Functional Requirements**
  - Quality attributes
  - Example: Performance, Security, Reliability

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### ◆ 3 Requirements Validation

#### ✓ Meaning:

Ensuring requirements are **correct, complete, and consistent**.

#### ✓ Why important?

To reduce errors before development starts.

#### ✓ Validation Techniques:

- Requirement reviews
- Prototyping
- Test case generation

#### ✓ Check for:

- Completeness
- Consistency
- Feasibility
- Verifiability

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## ◆ 4 Requirements Management

### ✓ Meaning:

Managing changes to requirements throughout the project.

### ✓ Why needed?

Requirements may change due to:

- Business changes
- New technology
- User feedback

### ✓ Activities:

- Requirement version control
- Change management
- Traceability
- Impact analysis

### ✓ Tool Support:

- JIRA
  - DOORS
  - Trello
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